The mRNA assembly line: transcription and processing machines in the same factory

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Processing of RNA precursors to their mature form often occurs co-transcriptionally. Consequently, the ternary complex of DNA template, RNA polymerase and nascent RNA chain is the physiological substrate for factors that modify the nascent RNA by capping, splicing and cleavage/polyadenylation. mRNA production is thought to occur within a 'factory' that contains the RNA polymerase II transcription machine and the processing machines. Newly discovered protein–protein contacts between RNA polymerase and factors that process mRNA precursors are beginning to illuminate how the 'mRNA factory' works.

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Abbreviations

CTD	carboxy-terminal domain
GT	guanylyltransferase
MT	7-methyltransferase
pol II	RNA polymerase II
pre-mRNA	messenger RNA precursor
SMN	survival of motor neurons
snRNP	small nuclear ribonucleoprotein
SR	serine/arginine-rich
TFIID/IIH	transcription factor IID/IIH

Introduction

Transcription of protein-encoding genes by RNA polymerase II (pol II) to make mRNA precursors (pre-mRNAs) is the primary step in gene expression but it is not permitted to take place undisturbed in isolation within the nucleus. Well before a full-length pre-mRNA can be completed, the nascent RNA is already modified by factors that work co-transcriptionally. In this review, I will focus on the coupling of pol II transcription with capping, splicing and cleavage/polyadenylation of mRNA precursors. Capping splicing and 3′ processing of pre-mRNAs can each occur in isolation *in vitro*; however *in vivo*, studies by electron microscopy (EM), crosslinking, and substituting other RNA polymerases for pol II indicate that most processing occurs co-transcriptionally rather than post-transcriptionally [1•,2••,3••,4–6]. Other nuclear processes that are known or suspected to occur co-transcriptionally and are not addressed in this review include mRNA packaging for export to the cytoplasm [1•,7], adenosine–inosine editing [8] translation [9,10] and DNA repair [11].

RNA polymerase II: a nexus of protein–protein interactions in the nucleus

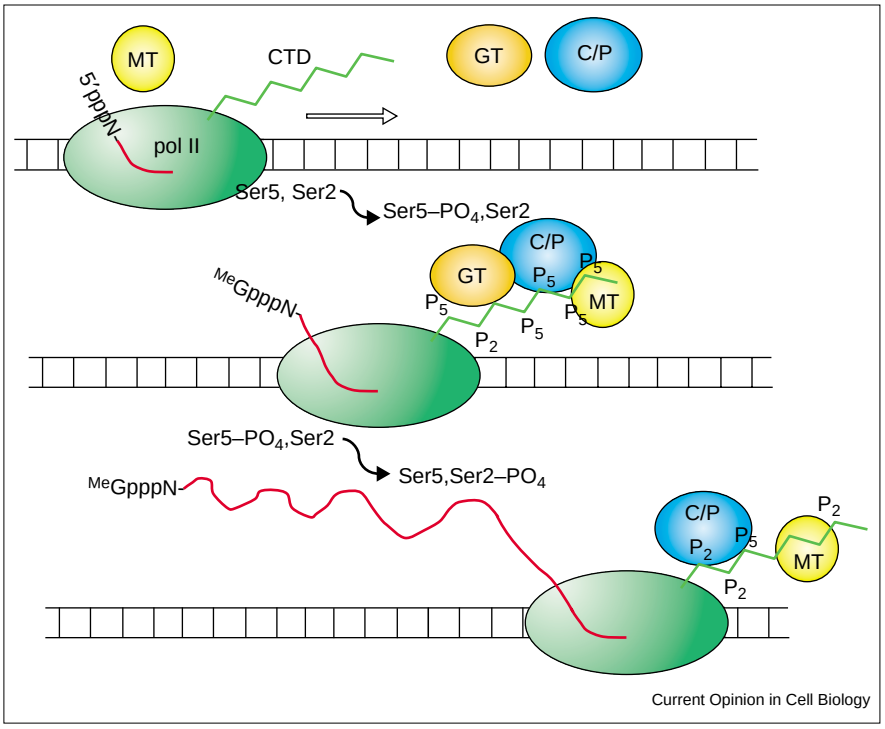
Protein-encoding genes are transcribed exclusively by pol II, and pre-mRNA processing is specifically targeted to transcripts made by this polymerase. Accordingly, pol II is especially equipped to cooperate with processing factors and other nuclear proteins, largely via interactions with a unique domain that protrudes from the large subunit of the enzyme [12]. This carboxy-terminal domain (CTD) of pol II is composed of tandem repeats (52 in vertebrates and 26 in budding yeast) of a heptad with the conserved consensus sequence YSPTSPS (single letter amino acid code). It is required for transcriptional activation and repression and for efficient capping, splicing and cleavage/polyadenylation of RNA transcripts [13–17]. Although these three RNA-processing steps are highly interdependent, the CTD stimulates each independently [18•] (see Update). The amino-terminal half of the human CTD supports capping but not splicing or 3′ processing, whereas the carboxy-terminal half supports all three processing steps [18•]. A short peptide sequence carboxy-terminal to the last heptad repeat is largely responsible for the different processing functions of the two halves of the CTD (N Fong and D Bentley, unpublished data).

The CTD appears to occupy a central position in the 'mRNA factory', where it functions as a platform for interaction with processing factors and other proteins. The CTD is more than a passive scaffold, however; indeed, it has been dubbed 'the tail that wags the dog' [19]. Binding to the CTD can regulate the activity of protein partners. Moreover, dynamic phosphorylation and dephosphorylation of the CTD on serine residues 2 and 5 of the heptads (Figure 1) during transcription appears to orchestrate an exchange of partners as the polymerase traverses a gene [2••,3••,20].

Capping: getting to the right place at the right time

The 7-methyl G5′ppp5′N cap is an identifying mark on all pol II transcripts. It is added when the RNA is about 25 bases long, soon after its 5′ end emerges from the RNA exit channel of the polymerase [21]. The cap is tacked on by three enzymes acting in sequence: RNA triphosphatase, guanylyltransferase (GT) and 7-methyltransferase (MT). How do these enzymes get to the right place at the right time to make their mark? GT and MT each bind exclusively to the phosphorylated CTD [13,22,23]. CTD phosphorylation by the general transcription factor TFIIH occurs shortly after initiation and is important for the transition to productive elongation. *In vivo* crosslinking experiments [2••,3••] in budding yeast showed that recruitment of all three capping enzymes to the 5′ ends of transcribed genes requires Kin28, the kinase subunit of TFIIH that phosphorylates Ser5. Moreover, the mammalian GT is activated allosterically by binding to Ser5-phosphorylated heptads [24]. CTD phosphorylation therefore serves multiple functions

Figure 1



Messenger RNA factory dynamics. The pol II CTD is phosphorylated predominantly on Ser5 shortly after initiation of the RNA chain. During elongation, there is a net loss of Ser5 phosphates (P5) and a net gain of Ser2 phosphates (P2). Capping enzymes guanylyltransferase (GT) and methyltransferase (MT), as well as cleavage/polyadenylation factors (C/P), are recruited at the 5′ end of the gene. During elongation, GT is selectively released and more C/Ps are progressively recruited.

early in the life of a pre-mRNA: it promotes elongation, recruits capping enzymes and allosterically activates the GT reaction.

What happens after the cap has been added? Are the capping enzymes jettisoned from the ternary complex or do they hang on and ride down the gene? The answer differs for different enzymes. The complex of RNA triphosphatase and GT is released early, in a step coupled with removal of Ser5 phosphates by Fcp1 [2••] or another phosphatase [20]. Unexpectedly, MT remains associated with pol II even at the 3′ end of the gene. The methyltransferase may travel with pol II for the whole length of the gene; or it could be bound and released in a dynamic equilibrium throughout elongation of the RNA chain. The association of MT with pol II well beyond the point when the cap has been added suggests that this enzyme has an additional function, perhaps in transcription.

There is a fascinating link between capping and transcription elongation and this may serve as a checkpoint to ensure that pol II commits to productive elongation only after the transcript has been capped. The cellular transcription factor Spt5 has both positive and negative effects on elongation; it also binds to GT and stimulates capping [25]. Furthermore, the HIV Tat transcription factor that stimulates transcription elongation by recruiting another CTD kinase, PTEFb, also binds GT and enhances capping *in vitro* [26•]. Tat also stimulates the extent of capping of HIV reporter gene transcripts *in vivo* (D Zorio, S McCracken, D Bentley, unpublished data). Factors such as Tat previously thought of as exclusively transcription factors now appear to also affect capping, both directly through contacts with capping enzymes and indirectly by enhancing CTD phosphorylation. The details of the capping/elongation connection remain to be clarified but communication between these factors could make sure that RNA does not get too long before the cap is added. The cap probably needs to be in place by the time the first intron is spliced, because the cap binding complex, CBC, stimulates removal of this intron. Splicing of this intron is enhanced by interaction between U1 small nuclear ribonucleoprotein (snRNP) and CBC, which, like other mRNA-binding proteins, associates with its target co-transcriptionally [1•].

Splicing and elongating RNA polymerase II

The connection between transcriptional elongation and RNA processing has been extended recently to include splicing as well as capping. Introns can be removed by the spliceosome within 30 seconds of transcription of the 3′ splice site [27••]; however, post-transcriptional splicing has also been documented [4,28]. The fraction of introns removed co-transcriptionally versus post-transcriptionally is not accurately known; nor is it known whether post-transcriptional splicing requires co-transcriptional marking of the intron. The spliceosome is a large, dynamic complex comprising five uridine-rich snRNPs (UsnRNPs) and many other proteins. The splicing UsnRNPs form a complex with the elongation factor TAT–SF1, which associates with pol II via the CTD kinase PTEFb [29••]. The UsnRNP–TAT–SF1 complex stimulates both *in vitro*

transcription and splicing. Moreover, a template with a functional intron was transcribed better than that with mutations in the 5′ or 3′ splice sites. The transcription/splicing connection is further cemented by the discovery of a form of pol II holoenzyme that includes UsnRNPs and serine/arginine-rich (SR) proteins [30], and by the finding that yeast Prp40, a U1 snRNP subunit, binds to the CTD [31].

Spliceosome assembly is stimulated by SR proteins that can bind to exonic splicing-enhancer elements co-transcriptionally [1•] and regulate constitutive and alternative splicing. The phosphorylated CTD enhances *in vitro* splicing by stimulating spliceosome assembly [32,33] and it is required for splicing enhancers to work *in vivo* [18•]. Furthermore, the CTD is necessary [34] and probably sufficient [17] for localization of transcription complexes to the periphery of nuclear speckles or interchromatin granules, structures where SR proteins and other mRNA processing factors are concentrated. Together, these experiments strongly suggest a connection between pol II and SR protein function in splicing.

Transcription and spliceosome recycling

Because most genes have multiple introns, splicing factors, unlike the capping and cleavage/polyadenylation factors, must be able to perform their job many times on a single growing RNA chain. After each intron is removed, the spliceosome is thought to disassemble. So, can pol II carry enough cargo to support assembly of multiple spliceosomes? This question was addressed by EM reconstruction of pol II transcription complexes on the *Chironomus BR3* gene [27••], which contains 38 introns. Several classes of transcription complex varying in shape and size between 3.9 and 6.3 MDa were distributed at intervals of about 0.5 kb over the entire gene. Immunostaining showed that these particles contained both pol II and U2 snRNP. Clearly, these ternary pol II complexes are not large enough to supply more than one spliceosome (estimated molecular weight: 4 MDa) at any given time. Therefore, spliceosomes must rapidly recycle, apparently within the context of a complex with pol II.

Perhaps one role of pol II is to nucleate protein assemblies that re-use spliceosome subunits at successive introns in a growing pre-mRNA. A candidate recycling factor associated with pol II is the SMN (survival of motor neurons) complex, which binds methylated spliceosomal Sm proteins and stimulates UsnRNP assembly (see Dreyfuss and colleagues, this issue [pp 305–312]). The SMN complex binds the pol II CTD indirectly, by association of the SMN protein with RNA helicase A [35•].

Promoters as determinants of mRNA fate

A surprising consequence of communication between the transcription and splicing machines is that promoters can control decisions between splicing at alternative sites that might be situated thousands of bases downstream [36]. The molecular basis for how promoters determine the fate of mRNAs is a major unanswered question. One suggestion is that modulation of CTD phosphorylation at the promoter could indirectly influence downstream processing by determining which factors bind the pol II elongation complex. Another possibility is that factors recruited to promoters are able to affect splicing directly. The latter possibility is supported by the discovery of a class of proteins that bind promoters like classical transcription factors and co-activators but also resemble splicing factors in having SR domains (for example, papillomavirus E2 and PGC1 [37,38•]) or making contacts with RNA or RNA-binding proteins (for example, DEK and PU.1 [39,40]). A number of such hybrid proteins have been created by tumor-associated chromosome translocations (for example, PSF–TFE3, TLS–ERG and EWS–Fli [41–43]). It is not clear whether these hybrid proteins remain bound to the promoter and act at a distance to affect splicing or whether they hop onto the polymerase when it departs and influence splicing locally.

Polymerization by pol II is relatively slow: 1000–2000 bases per minute compared with 15,000 bases per minute for T7 RNA polymerase. Sluggish elongation may be necessary to permit co-transcriptional processing, just as slow elongation in *Escherichia coli* is necessary for co-transcriptional translation. Splicing *in vitro* was more efficient when coupled to transcription by pol II than to T7 RNA polymerase [44]. Promoter-bound transcriptional activators that modulate elongation affect alternative splice-site choice in the fibronectin gene [45•]. This finding suggests, intriguingly, that promoters can pre-ordain the fate of pre-mRNAs by influencing the rate of RNA-chain growth.

3′ Processing and the end of the road

How does pol II know when it has reached the end of the road? At the end of a gene, the pre-mRNA is cleaved at the poly(A) site by an unidentified endonuclease, and a poly(A) tail is added to the exposed 3′ end. Termination, the release of pol II from the DNA template, occurs at diffuse positions hundreds of bases downstream of the poly(A) site. Termination requires a functional poly(A) site, but not cleavage of the RNA [46,47••]. This requirement makes satisfying sense, as it ensures that termination occurs after the 3′ end of the gene has been transcribed. The mechanism underlying poly(A)-site-dependent termination remains an important problem, but recently developed *in vitro* termination systems [47••,48•] and the discovery of protein–protein contacts between 3′ processing factors and pol II promise to clarify this question in the near future. Transcriptional pause sites and RNA cleavage sites downstream of the poly(A) site may be important for 3′ processing and termination at some genes [48•,49].

In yeast, termination requires a subset of cleavage/polyadenylation factors, including Rna14, Rna15 and Pcf11 [50] and the peptidyl-prolyl isomerase Ess1 [51]. In mammalian cells, the CTD is required [14]. Direct contacts

Table 1

Proteins implicated in both mRNA processing and pol II transcription and/or transcription-coupled DNA repair.

Factor	Capping	Splicing	3′-end processing	pol II binding	Transcription initiation/ elongation	Transcription termination	Chromatin remodelling	DNA repair	Reference
SAP130		+					+	+	[61]
PGC1		+		+	+				[38•]
Tat	+			+	+				[26•]
Spt5	+			+	+				[25]
GT	+			+					[21]
MT	+			+					[21]
U snRNPs		+			+				[29••,30]
WT1		+			+				[62]
PU.1		+			+				[40]
EWS–Fli		+		+	+				[43]
PSF–TFE3		+			+				[41]
p54^{nrb}–TFE3		+		+	+				[41]
SMN		+		+					[35•]
TLS-ERG		+			+				[43]
DEK		+			+				[39]
E2		+			+				[37]
SR proteins		+		+					[30]
Aly/REF		+			+				[63]
Prp40		+		+					[31]
Ess1			+	+	+	+	+		[51,54,64]
CstF50			+	+				+	[11,18•]
CPSF			+	+	+				[14,60]
PC4/Sub1			+		+	+			[56•]
MBP1			+		+	+			[58•]
p52		+			+				[65]
CFII(m)		+	+		+			+	[66]

Note: in many cases the precise functions served by these proteins in multiple nuclear events are not known but are inferred from protein–protein interactions.

have been found between the CTD of pol II and 3′ processing factors, including mammalian CstF p50 [18•] and yeast Pta1 [52], Rna14, Pcf11 [53] (see Update) and Ess1 [54]. Another yeast CTD-binding protein, Nrd1, is required for termination at small nucleolar RNA (snoRNA) genes that encode nonpolyadenylated transcripts [55••]. The carboxyl termini of the homologous 3′-end processing factors Rna15, CstF64 and Cft1 make additional contacts with the co-activator PC4/Sub1 [56•], which may inhibit CTD phosphorylation [57], and the cell-cycle-regulated transcription factor Res2/MBP1[58•], which has been implicated in triggering termination. Exactly how all these interacting proteins work to achieve poly(A)-site-dependent termination is still unclear.

Several recent studies show that yeast and *Xenopus* 3′ processing factors contact elongating pol II transcription complexes along the entire length of genes ([3••,59]; see also Update). The presence of 3′ processing factors at the 5′ ends of genes seems surprising but it is supported by the fact that cleavage polyadenylation specificity factor (CPSF) can be recruited to promoters in association with TFIID and may then be handed off to the polymerase [60]. One model suggests that termination is triggered by a change in the interaction between 3′ processing factors and polymerase after it passes the poly(A) site, which flips a switch to non-processive transcription [14], perhaps by loosening the jaws with which pol II clamps down on the DNA. Interestingly, the CTD is located close to the flap domain of the pol II second largest subunit that forms part of this clamp [12].

Conclusions and future prospects

Despite the numerous links that have been discovered between so-called transcription factors and processing factors (see Table 1), the inside of the mRNA factory remains a mystery. It is also not known to what extent the complexes between the transcription and processing machinery are pre-assembled, perhaps in specialized workshops (Cajal bodies) [59], as opposed to being newly assembled on the gene in each round of transcription. Several 3′ processing factors, capping enzymes and UsnRNPs travel as passengers with pol II, either transiently or permanently, as it traverses a gene. How the composition of the pol II ternary complex changes as it transcribes through exons, introns and 3′-end processing sites remains an important challenge for the future. It seems clear, though, that the distinction between transcription factors and RNA-processing factors will continue to fade.

The myriad of connections between different nuclear machines appears to constitute the structural basis for an extensive network of communication that serves not just to stimulate co-transcriptional RNA processing but also

to influence transcription, mRNA export and possibly other processes, such as nuclear translation, DNA repair and replication. A challenge for future work in this field is to decipher the signals being transferred through this network.

Update

Recent work has shown that cleavage at a poly(A) site uncoupled from transcription is stimulated by the CTD, that this activity requires at least 26 heptad repeats and that 52 consensus heptads (YSPTSPS) work as well as the natural CTD [67].

It has also been found that in yeast, subunits of the CFI and PFI 3′ processing factors associate with pol II at the promoter and are progressively recruited as polymerase travels along the gene. Furthermore, Pcf11 was found to bind specifically to CTD heptads phosphorylated at the Ser2 position [5]; however, the CTD is not as critical for 3′ processing and splicing in yeast as it is in mammalian cells. Reference 68 is an excellent recent review of the field.

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